

Know-your-own-League context: insights for player preparation and recruitment - Part 3: Player substitutions

Martin Buchheit,^{1 2 3} Maxime Settembre,¹ Karim Hader,¹ Adrien Tarascon,² Derek McHugh,¹ Raymond Verheijen⁴

¹Kitman Labs, Dublin, Ireland

²Lille OSC, Lille, France

³HIITscience, Revelstoke, Canada

⁴Football Coach Evolution, The Netherlands

Association football | Team performance | Home and away | Line-up changes | Goals scored

Headline

It's often said that while substitutions can solve a problem, they can also introduce a new one (7). In fact, while maintaining the same quad on the pitch for 90 min. can be seen as optimal for overall match performance when things go well (i.e., which favours better between-player communication and tactics), substitutions are often necessary to maintain game pace when players are getting tired over the course of a match. For these reasons, deciding when and how many players to substitute during a match is a key question for every manager (4, 7). Following our work on team formations (2) and results and goals scored (3), this third part in our series of papers focuses on player substitutions over the past two decades in the top seven European leagues.

Aim

This study aims to examine the between-league differences in player substitutions. These were all examined for both home vs. away matches.

Methods

Data extraction

We extracted the fixtures data of the teams competing in the top seven European leagues i.e., England, France, Germany, Holland, Italy, Portugal and Spain. This includes 21 seasons from 2001/02 to 2021/22. Only league fixtures were considered for this part. This data was scraped from Transfermarkt. Across the 21 seasons, this represents 7 competitions, 269 teams and more than 50,000 fixtures.

Data Analysis

To display match substitutions, we used boxplots to convey the spread across teams within the league. Boxplots display the median, with the size of the box (Interquartile range, IQR) spanning from Q1 (25th percentile) to Q3 (75th percentile). The outside bars are the lower and upper whiskers: the lower whisker is the smallest value in the dataset that is no smaller than $Q1 - 1.5 \times IQR$ and the upper whisker is the largest value that is no greater than $Q3 + 1.5 \times IQR$. Values outside of both whiskers are considered outliers and are not shown here. We also compared the proportion of teams' goal difference and total number of goals scored in a game when teams used a low (i.e., 0 or 1) and the maximum (i.e., 3) number of substitutions. For this analysis specifically, only the observations before the start of COVID were used; therefore, the maximum

number of substitutions by a team during a match is three (and not five as currently, after the COVID period). Additionally, only the observations (team during a match) with complete substitution data were considered. Observations with substitutions due to injuries were excluded from this analysis. In total, there were 12501 observations with three substitutions, 557 observations with only 1 substitution, and 344 observations without substitution at all.

Discussion

This study is the first to our knowledge to examine between-league differences in players' substitutions.

The main limitation of the current analysis remains the data set obtained from an online database. In particular, there is always a degree of uncertainty in the way a substitution is recorded on a public website like Transfermarkt relative to what was actually registered.

The team-average number of substitutions per match (home and away matches) per season over the past two decades is shown for each league individually in Figures 1 and 2 (ranked based on UEFA Index, 26/10/2022). There was also no clear difference in the observed trends between home and away matches. Our main results are that in terms of player substitutions, things have remained stable across the years and there doesn't seem to be an effect of match location either.

Until the 2018/19 season, the number of substitutions per team tended to be similar, i.e., close to three for all leagues, and clearly stable across time. Following FIFA new regulation implemented in most leagues after the COVID pandemic (6), substitution numbers then moved up to four to five in all leagues, except in the EPL, where substitution numbers remained around three - as per the league-specific regulation (Figures 1 and 2). The fact that coaches have consistently used the highest number of substitutions allowed (three and then five) also suggests that substitutions are a complete part of match performance, allowing teams to 1) bring fresh legs on the pitch, but also 2) adjust tactics as the match advances.

It is however worth noting that before 2018/19, boxplot IQRs and whiskers (Figures 1 and 2) are quite small due to all teams likely having a similar strategy in terms of the number of substitutions; but with the new five-sub rule (2019/20 onwards), the spread across teams becomes greater, suggesting that some teams may not have a squad depth that enables them to substitute five players per match, which may also increase the gap between top teams and the others (Buchheit 2022, O'Hanlon 2020).

We then compared the proportion of teams' goal difference (Figure 3) and total number of goals scored in a game (Figure 4) when teams used a low (i.e., 0 or 1) and the maximum (i.e., 3) number of substitutions.. There was a trend for a higher proportion of observations with low substitutions when the goal difference was equal to zero (i.e., draw, Figure 3), especially in the case of 0-0 or 1-1 (Figure 4). While it's difficult to draw conclusions without more information about the context (i.e., players available, line-up and match dynamic), this scenario may reflect coaches' conservative attitudes during tight matches, when they may be reluctant to change their team composition while waiting for the scoreline to develop (e.g., fear of not seeing the substitute adapt to the match context, tactical paralysis). It's also difficult to know whether substitutions are the cause or the effect of large victories or losses, but Figures 3 and 4 also suggest that more goals are scored and clearer outcomes are shown when three substitutions are made.

Key findings

- Up until the season 2018/2019, the number of substitutions per team (i.e., close to three, Figures 1 and 2) tended to be similar between all leagues, and clearly stable across time.
- Substitution numbers then moved up to 4.5 after the COVID pandemic, except in the EPL, where the substitution number remained around three (as per the league-specific regulation).
- There was no clear difference between home vs. away matches in terms of substitutions.
- Overall the data suggest that when coaches can make substitutions, they do so. However there appears to be some small effect of scoreline results and match dynamic that may affect the actual number of substitutions.
- There was a trend for a higher proportion of observations with low (i.e., 0 or 1) vs. maximum (i.e., 3) substitutions numbers when the goal difference was equal to zero (i.e., draw), especially in the case of 0-0 or 1-1 scores.

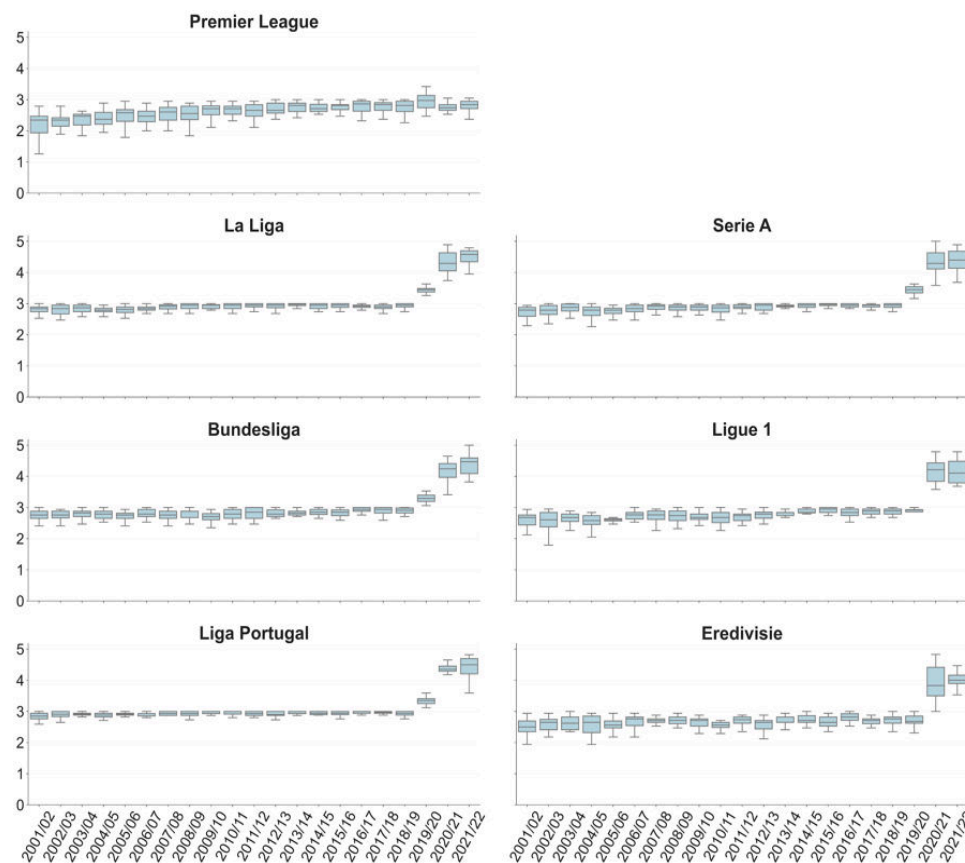


Fig. 1. Team-average number of substitutions per home match per season in the seven leagues over the past two decades.

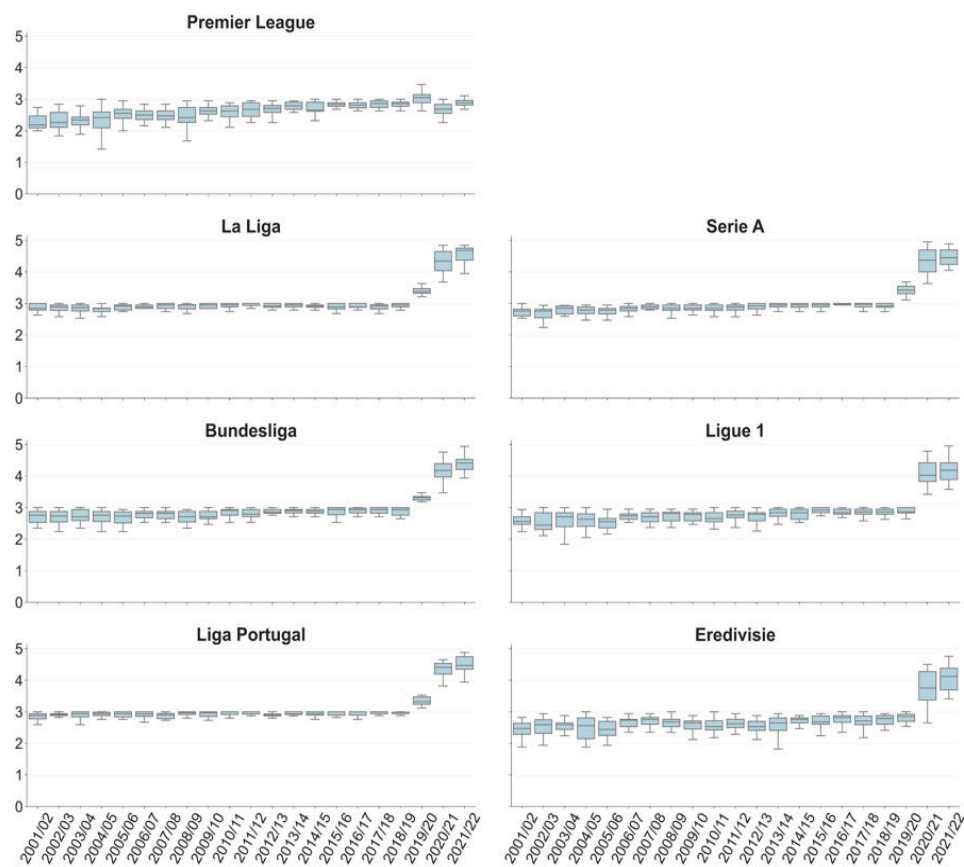


Fig. 2. Team-average number of substitutions per away match per season in the seven leagues over the past two decades.

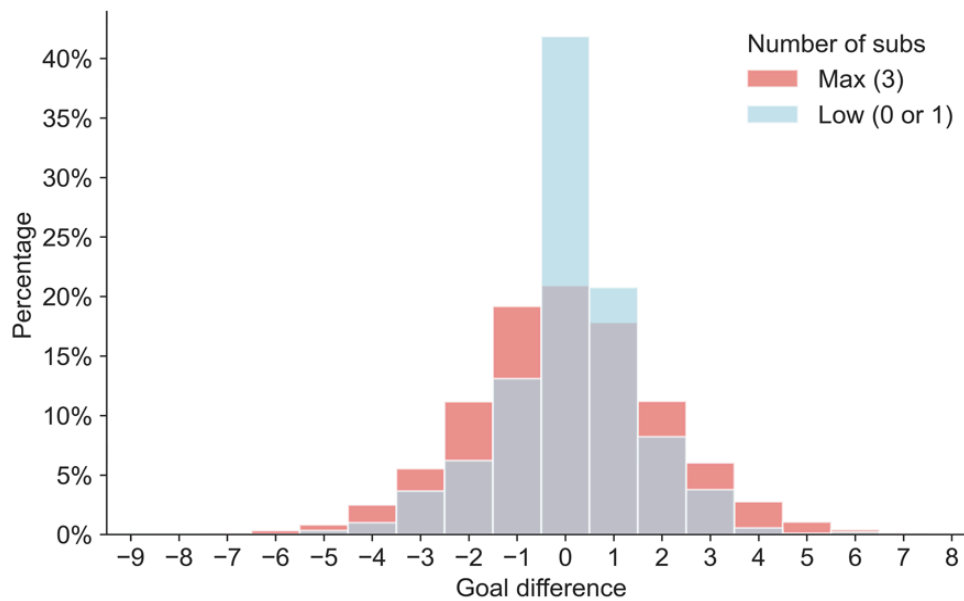


Fig. 3. Proportion of teams' goal difference when their number of substitutions is the highest (3) vs. when it is low (0 or 1)

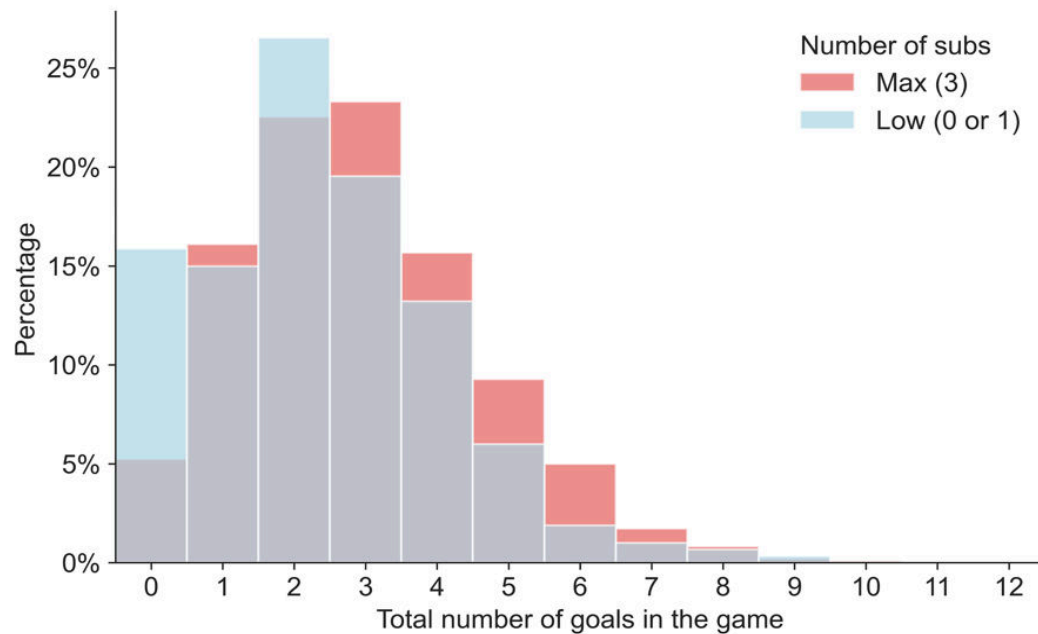


Fig. 4. Proportion of the total number of goals in a game when the number of substitutions of one of the two teams is the highest (3) vs. when it is low (0 or 1)

Twitter: Martin Buchheit (@mart1buch), Karim Hader (@Karad70), Adrien Tarascon (@TarasconAdrien), Derek McHugh (@DerekMcHugh3), Raymond Verheijen (@raymondverheije)

References

1. Buchheit M, Settembre M, Hader K, Tarascon A, McHugh D & Verheijen R. Do mid-week European matches influence European teams' performance in their domestic league? A 20-year study. *Sport Perf & Science Reports*, Oct 22, 175, v1
2. Buchheit M, Settembre M, Hader K, Tarascon A, McHugh D & Verheijen R. Know-your-own-League context: insights for player preparation and recruitment - Part 1: Team formations. *Sport Perf & Science Reports*, March 03, 181, v1.
3. Buchheit M, Settembre M, Hader K, Tarascon A, McHugh D & Verheijen R. Know-your-own-League context: insights for player preparation and recruitment - Part 2: Results and goals scored. *Sport Perf & Science Reports*, March 10, 183, v1.
4. Clayfield, B. Number of Substitutions in Soccer (And Reason for the Limit). Yoursoccerhome blog, accessed Feb 28, 2023.

son for the Limit). Yoursoccerhome blog, accessed Feb 28, 2023.

5. O'Hanlon. Premier League and squad depth: Is rotation really an issue for Liverpool, Man City & Co? *ESPN*. Jan 23, 2020.

6. The International Football Association Board (IFAB). www.ifab.com. Law 3.2

7. Verheijen R. Tactical principles. Methodological steps within tactical principles during team building process. *Football Coach Evolution*. June 2022.

Copyright: The article published on *Sport Performance and Science Reports* are distributed under the terms of the Creative Commons Attribution 4.0 International License (<http://creativecommons.org/licenses/by/4.0/>), which permits unrestricted use, distribution and reproduction in any medium, provided you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made. The Creative Commons Public Domain Dedication waiver (<http://creativecommons.org/publicdomain/zero/1.0/>) applies to the data made available in this article, unless otherwise stated.

